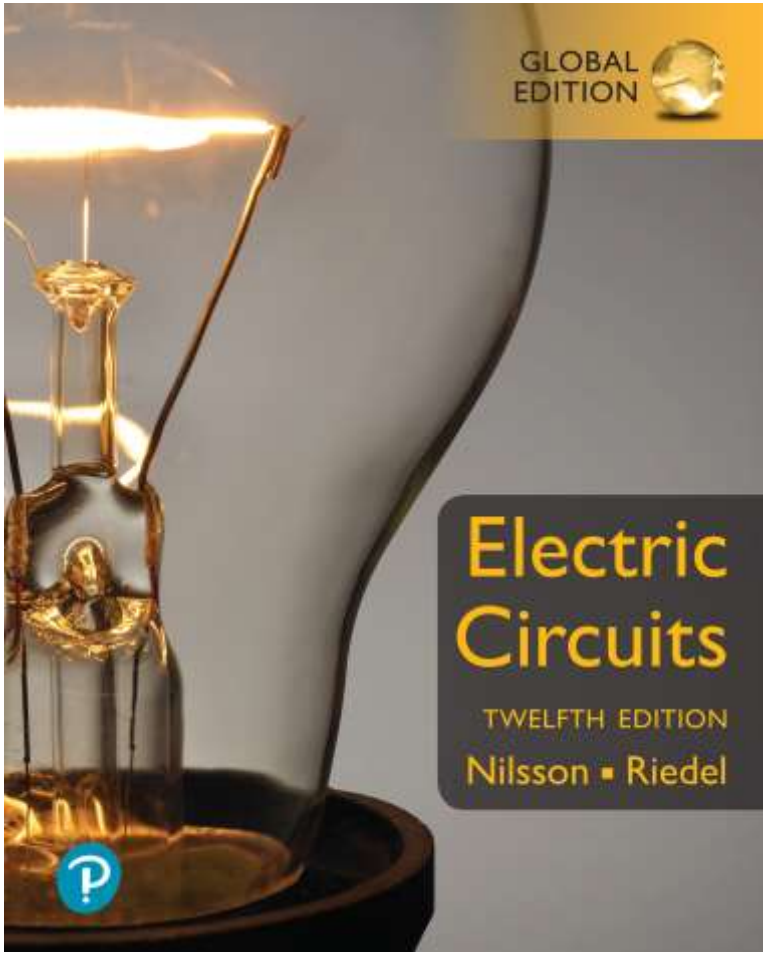
Electric Circuits

Twelfth Edition, Global Edition



Chapter 4 – Part 7

Techniques of Circuit Analysis

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Chapter 4 – techniques of circuit analysis (1 of 17)

Thevenin and Norton equivalents

- A circuit simplification technique
- Can be applied to any circuit consisting of resistors and sources



- Finding the Thevenin equivalent is a two-step process:
  - Find the open-circuit voltage, V_{oc}
  - Find the short-circuit current, i_{sc}

Then, $V_{Th} = V_{oc}$ and $R_{Th} = V_{oc} / i_{sc}$

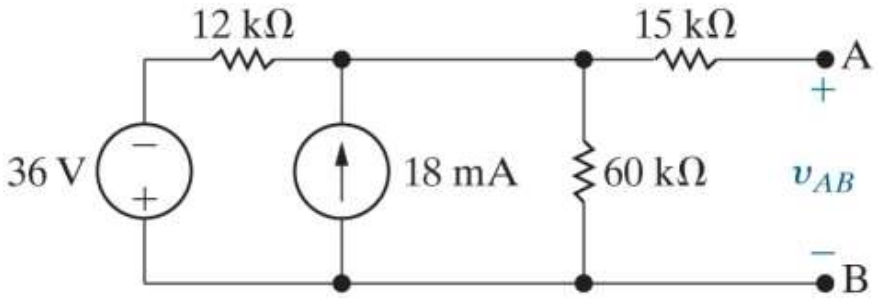
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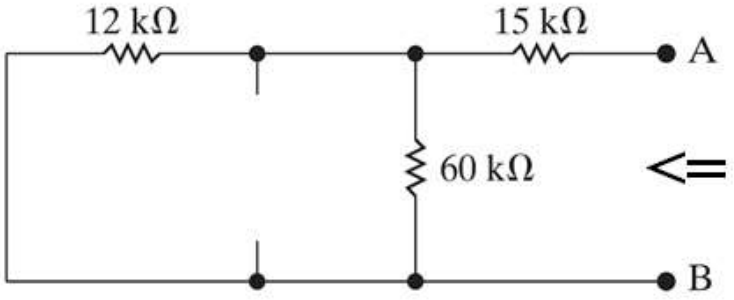
Chapter 4 – techniques of circuit analysis (2 of 17)

Thevenin and Norton equivalents

- Find the Norton equivalent of the circuit shown here.



Find R_{Th} :



$$R_{Th} = 15 + (60 \,||\, 12)$$

$$= 15 + 10 = 25\Omega$$

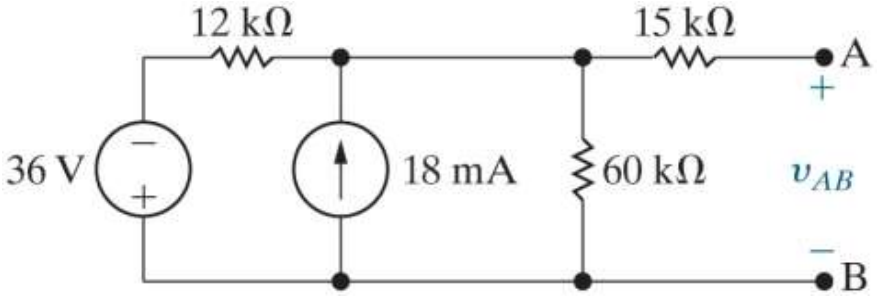
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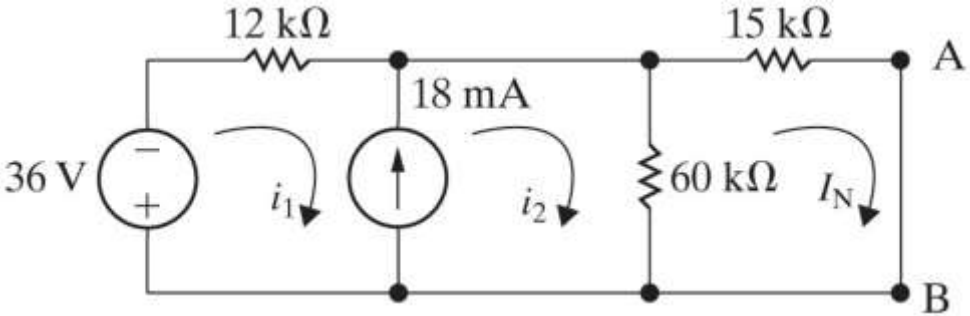
Chapter 4 – techniques of circuit analysis (3 of 17)

Thevenin and Norton equivalents

- Find the Norton equivalent of the circuit shown here (continued).



Find I_N using the mesh current method:



supermesh: $36 + 12{,}000i_1 + 60{,}000(i_2 - I_N) = 0$

I_N mesh: $15{,}000I_N + 60{,}000(I_N - i_2) = 0$

SM constraint: $i_2 - i_1 = 0.018$

Solving: $I_N = 6$ mA

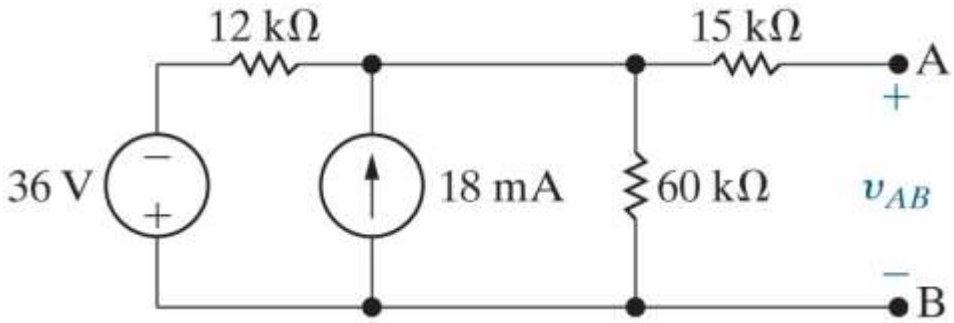
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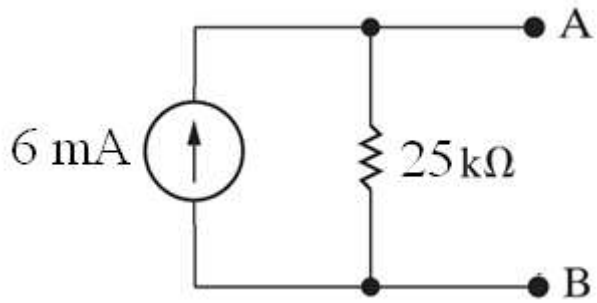
Chapter 4 – techniques of circuit analysis (4 of 17)

Thevenin and Norton equivalents

- Find the Norton equivalent of the circuit shown here (continued).



Draw the Norton equivalent:



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Chapter 4 – techniques of circuit analysis (5 of 17)

Thevenin and Norton equivalents

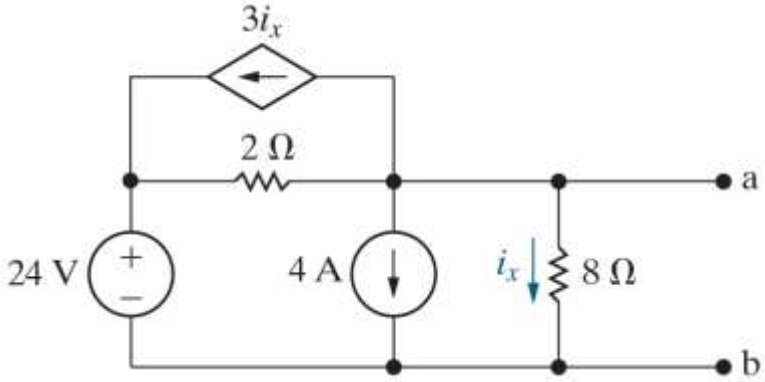
- Special cases:
  - Does the circuit have a dependent source? If so, **you must calculate the open-circuit voltage and the short-circuit current,** because you cannot “kill” the independent source(s) and make series and parallel combinations of resistors to find the Thevenin resistance.
  - Does the circuit have dependent sources but no independent sources? If so, the Thevenin voltage and Norton current are both zero! You have to calculate the Thevenin resistance using the test source method.

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Chapter 4 – techniques of circuit analysis (6 of 17)

If we use the node voltage method to find the open-circuit voltage, how many KCL equations will we write?



- ✗ A. 0
- ✓ B. 1
- ✗ C. 2
- ✗ D. 3

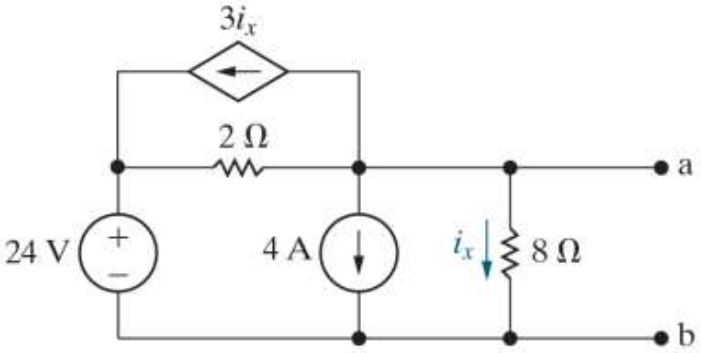
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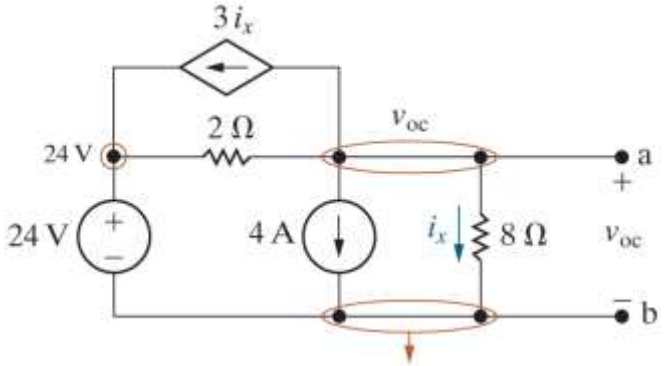
Chapter 4 – techniques of circuit analysis (8 of 17)

Thevenin and Norton equivalents

- Find the Thevenin equivalent to the left of the terminals a,b in the circuit below.



Find V_{OC} using the node voltage method:



At v_{oc}: $\frac{v_{oc}-24}{2}+4+\frac{v_{oc}}{8}+3i_x=0$

d.s. constraint: $i_x=\frac{v_{oc}}{8}$

Solving: $v_{oc}=8\ \text{V}$

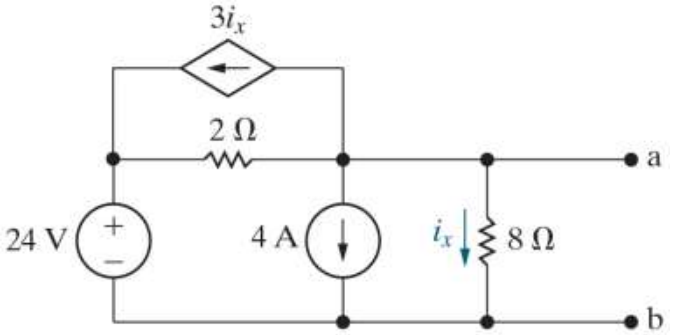
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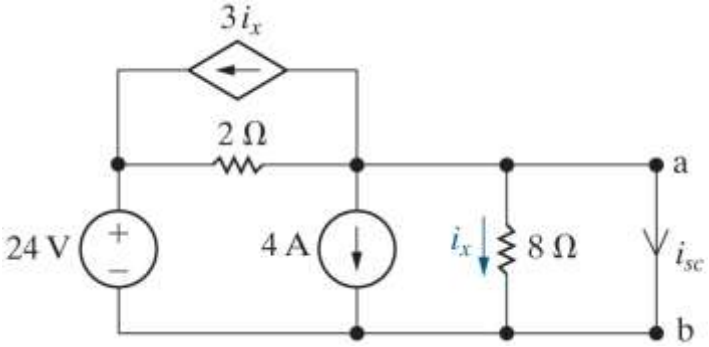
Chapter 4 – techniques of circuit analysis (9 of 17)

Thevenin and Norton equivalents

- Find the Thevenin equivalent to the left of the terminals a,b in the circuit below.



Find i_{sc}:



The short circuit eliminates the branch with the $8\ \Omega$ resistor from the circuit, which sets $i_x = 0$ and eliminates the dependent source. The simplified circuit is shown and analyzed on the next slide.

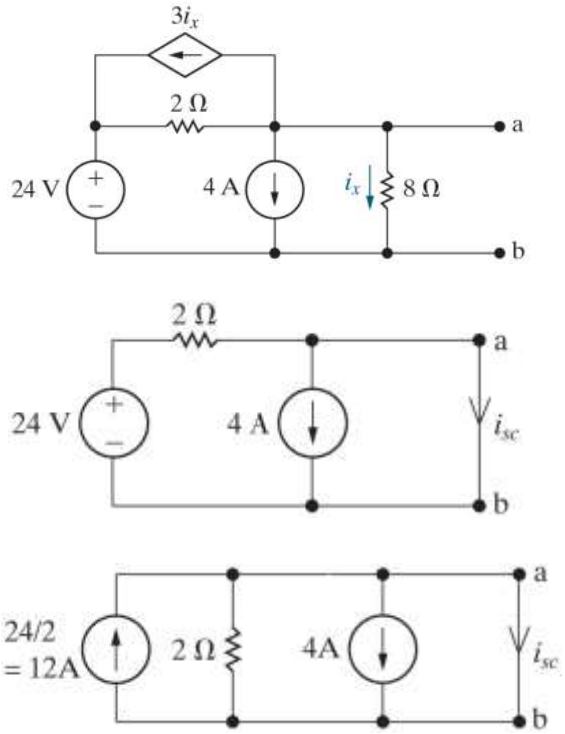
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Chapter 4 – techniques of circuit analysis (10 of 17)

Thevenin and Norton equivalents

- Find the Thevenin equivalent to the left of the terminals a,b in the circuit below (continued)



Find i_{sc} (continued):

Source transform this circuit to get:

The sum of the currents flows in the short circuit, which bypasses the branch with the 2 Ω resistor. Thus, the short circuit current is 12 – 4 = 8 A.

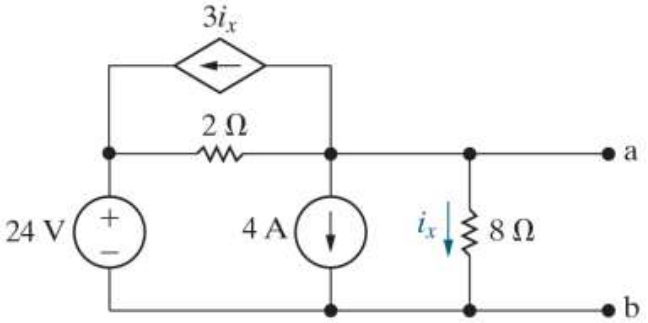
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Chapter 4 – techniques of circuit analysis (11 of 17)

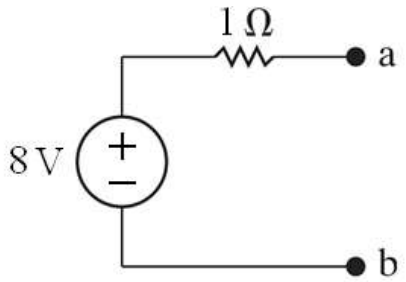
Thevenin and Norton equivalents

- Find the Thevenin equivalent to the left of the terminals a,b in the circuit below (continued)



Find R_{Th} and draw the Thevenin equivalent circuit:

$$R_{Th} = \frac{v_{oc}}{i_{sc}} = \frac{8}{8} = 1\,\Omega$$

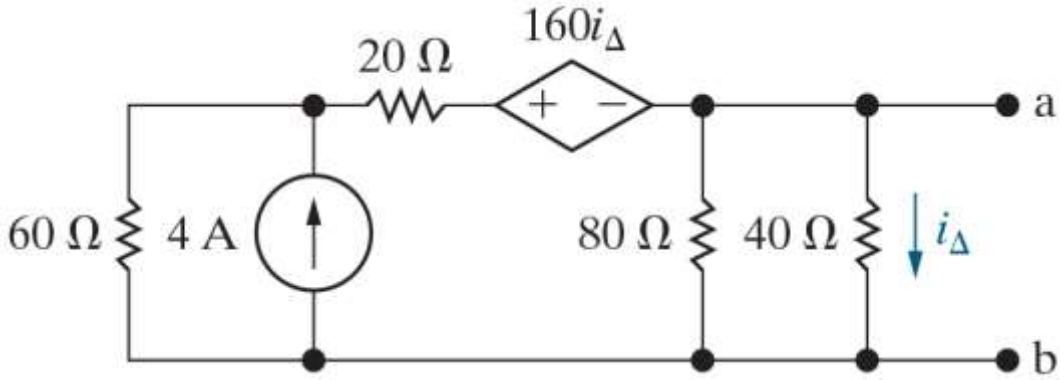


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Chapter 4 – techniques of circuit analysis (12 of 17)

By using the test source method on this circuit, we will be able to find



- ✔ A. The Thevenin equivalent resistance
- ✘ B. The open circuit voltage
- ✘ C. The short circuit current
- ✘ D. All of the above

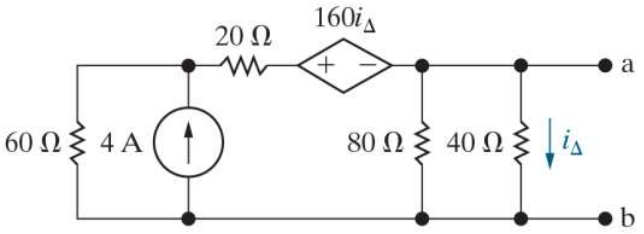
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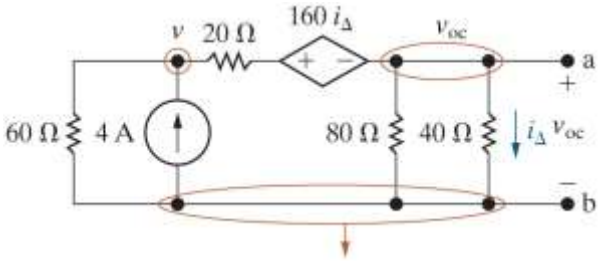
Chapter 4 – techniques of circuit analysis (13 of 17)

Thevenin and Norton equivalents

- Find the Thevenin equivalent to the left of the terminals a,b in the circuit below. Use the test source method.



Find V_{OC} using the node voltage method:



$$\text{At } v: \quad \frac{v}{60} - 4 + \frac{v - (160 i_\Delta + v_{oc})}{20} = 0$$

$$\text{At } v_{oc}: \quad \frac{(160 i_\Delta + v_{oc}) - v}{20} + \frac{v_{oc}}{80} + \frac{v_{oc}}{40} = 0$$

d.s. constraint: $i_\Delta = \dfrac{v_{oc}}{40}$ Solving: $v_{oc} = 30$ V

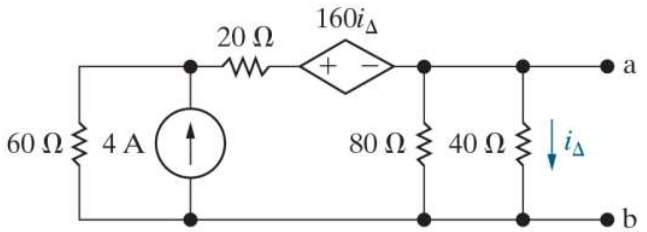
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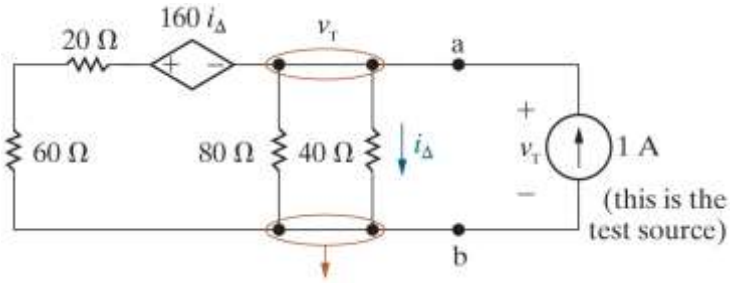
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Chapter 4 – techniques of circuit analysis (14 of 17)

Thevenin and Norton equivalents

- Find the Thevenin equivalent to the left of the terminals a,b in the circuit below. Use the test source method(continued)





Find R_{Th} **using the test source**

<u>method</u> – don't forget to kill the independent sources!

$$\text{At } v_T: \quad \frac{v_T + 160 i_\Delta}{280} + \frac{v_T}{60} + \frac{v_T}{60} - 1 = 0$$

$$\text{d.s. constaint: } i_\Delta = \frac{v_T}{40} \qquad \text{Solving: } v_T = 10 \text{ V}$$

$$\therefore \quad R_{Th} = \frac{v_T}{i_T} = \frac{10}{1} = 10\Omega$$

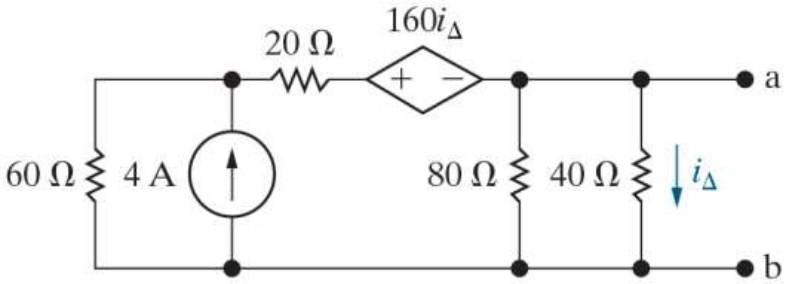
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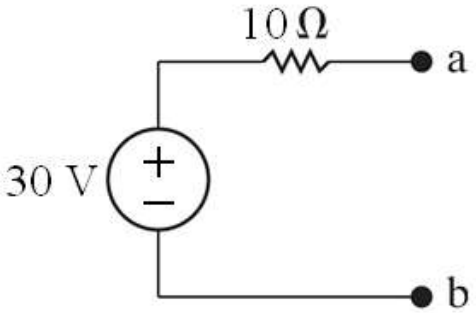
Chapter 4 – techniques of circuit analysis (15 of 17)

Thevenin and Norton equivalents

- Find the Thevenin equivalent to the left of the terminals a,b in the circuit below. Use the test source method (continued)



Draw the resulting Thevenin equivalent circuit:



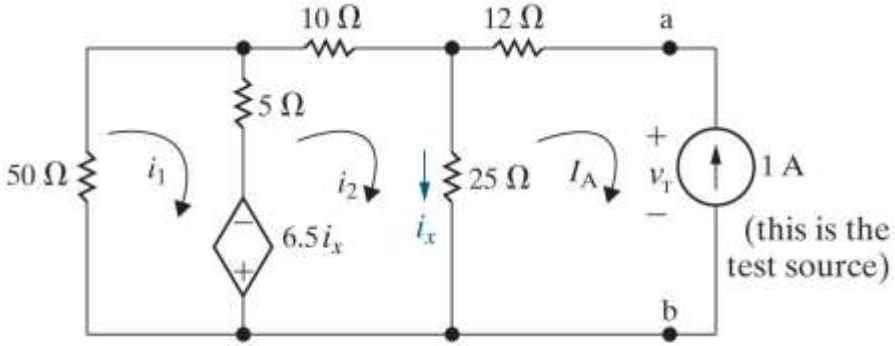
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Chapter 4 – techniques of circuit analysis (16 of 17)

Thevenin and Norton equivalents

- Find the Thevenin equivalent to the left of the terminals a,b in the circuit below. Use the test source method.



i_1 mesh: $50i_1 + 5(i_1 - i_2) - 6.5i_x = 0$

i_2 mesh: $10i_2 + 25(i_2 + 1) + 6.5i_x + 5(i_2 - i_1) = 0$

d.s. constraint: $i_x = i_2 + 1$

Solving: $i_1 = -24$ mA, $i_2 = -680$ mA, $i_x = 320$ mA

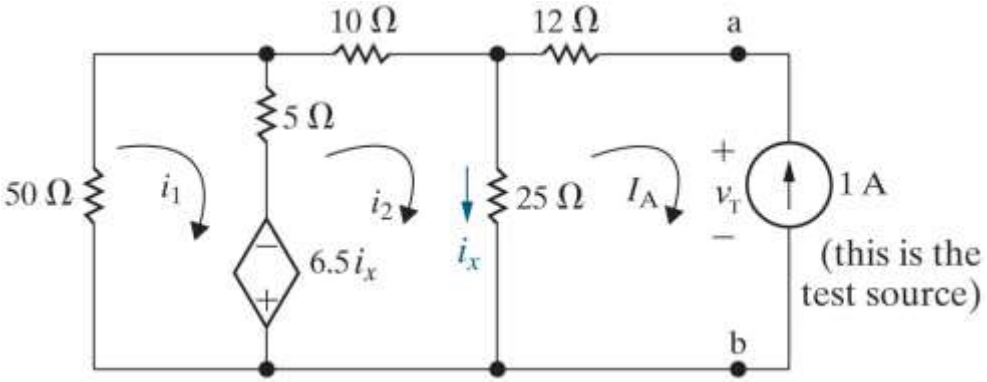
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Chapter 4 – techniques of circuit analysis (17 of 17)

Thevenin and Norton equivalents

- Find the Thevenin equivalent to the left of the terminals a,b in the circuit below. Use the test source method (continued)



KVL left mesh: $12(-1)+v+25(-1+0.68)=0$

$\therefore \quad v = 20 \text{ V}$

$\Rightarrow \qquad R_{Th} = \dfrac{20 \text{ V}}{1 \text{ A}} = 20\Omega$

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